

CASE 13

X-LINKED LYMPHOPROLIFERATIVE SYNDROME

A defect in the immune response to a virus.

Viruses pose a constant challenge to our immune system. Unable to reproduce on their own, viruses have evolved as parasites, capable of residing within living cells whose biosynthetic machinery they subvert for their own reproduction. The human immune system combats these hidden, invading microbes through a two-pronged strategy: 1) elimination of viruses before they are able to enter host cells and 2) identification and destruction of infected host cells that harbor viruses. The innate and adaptive immune systems are both enlisted in the defense against viruses.

Innate pattern recognition receptors, including the endosomal toll-like receptors (TLRs) 3, 7, 8, and 9 along with cytosolic RIG-like receptors (RLRs), sense intracellular viral particles by identifying prokaryotic nucleic acid fragments within the host cell. Activated TLRs and RLRs induce signaling through the interferon regulatory factor (IRF) family of transcription factors to produce type I interferons (most importantly, $IFN\alpha$ and $IFN\beta$). These interferons engender an antiviral state within the host by promoting transcription of genes that resist viral infection, increasing lymphocyte retention in the lymph nodes, favoring T_H1 differentiation, and enhancing the cytotoxic effector functions of natural killer (NK) and CD8 T cells.

NK cells of the innate immune system also combat viral infections through the cytotoxic response, eliminating infected host cells (Fig. 13.1). NK cells are large granular cells of the lymphocyte lineage; however, they differ from the adaptive immune system's B and T cell lymphocytes in that they express only germline DNA receptors. Hence, these cells are equipped to react immediately upon the first encounter with the virus and do not require activation, clonal expansion, and differentiation into effector cells like the cytotoxic T lymphocytes (CTLs). NK cells constantly survey host cells with telltale markers of viral infection. NK cells express receptors that activate or inhibit the cell's killer activity. The fate of potential target cells is determined by the balance of activating and inhibitory signals delivered

TOPICS BEARING ON THIS CASE:

NK-cell activating receptors and killer function

NK T cells

Cell signaling from tyrosine kinase-associated receptors

Activation and function of cytotoxic T cells

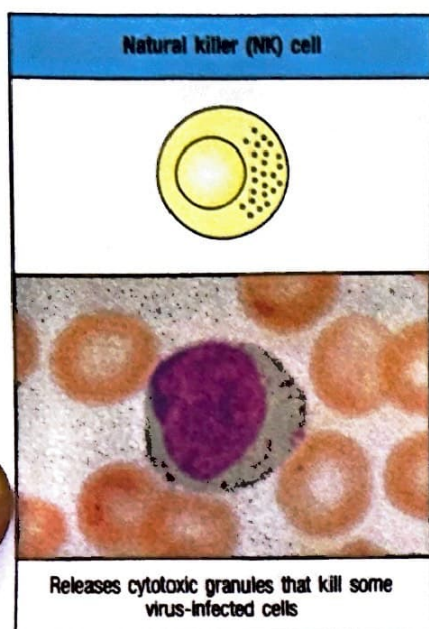


Fig. 13.1 Natural killer (NK) cells. These are large granular lymphocyte-like cells with important functions in innate immunity. Although lacking antigen-specific receptors, they can detect and attack certain virus-infected cells. Photograph courtesy of N. Rooney and B. Smith.

to NK cells through these receptors (Fig. 13.2). Inhibitory receptors interact with MHC class I molecules and help prevent NK cells from attacking healthy, uninfected cells. Once infected host cells are recognized, NK cells execute a cytotoxic response similar to CTLs, releasing granules that contain perforin and granzymes that induce apoptosis in the virus-infected cell (see Case 14). INF γ , which leads to macrophage activation and polarization of naive T cells to T_H1 cells, is also produced by NK cells.

In the adaptive immune response, virus-specific CTLs are generated during the primary immune response. A naive virus-specific CD8 T cell is activated to CTL effector status when its T cell receptor (TCR) recognizes a viral peptide loaded on a major histocompatibility complex (MHC) I molecule. Naive CD8 T cells are not easily activated to effector status. Strong co-stimulation by the dendritic cell together with help from activated CD4 T cells is usually required to activate a naive cytotoxic T cell. Once activated, CTLs kill infected cells through the release of cytotoxic granules similar to those of NK cells. Additionally, CTLs can induce cell death in a granule-independent manner via the Fas ligand, which engages the death receptor Fas on a host cell and triggers apoptosis. During the primary immune response to viruses, memory cytotoxic T cells are also produced. In the event of re-exposure to the same virus, either by reinfection from the environment or by reactivation of virus latent in the body, these cytotoxic T cells rapidly recognize and kill infected host cells displaying viral antigens.

In addition to CD8 T cells, another population of T lymphocytes, NKT cells, is thought to be important in the defense against viral infections. NKT cells share features found in both innate NK cells and T cells. They express markers, like CD56, which are found on NK cells. However, unlike NK cells, NKT cells have TCRs. These TCRs do not recognize peptides; instead, they identify lipid moieties that are loaded on an MHC-I-like molecule called CD1.

B cells, like CTLs and NKT cells, are also important to antiviral immunity. Neutralizing virus-specific primary and secondary antibodies often prevent viral entry into host cells. Further, antibodies coat viruses, opsonizing the microbe to promote clearance.

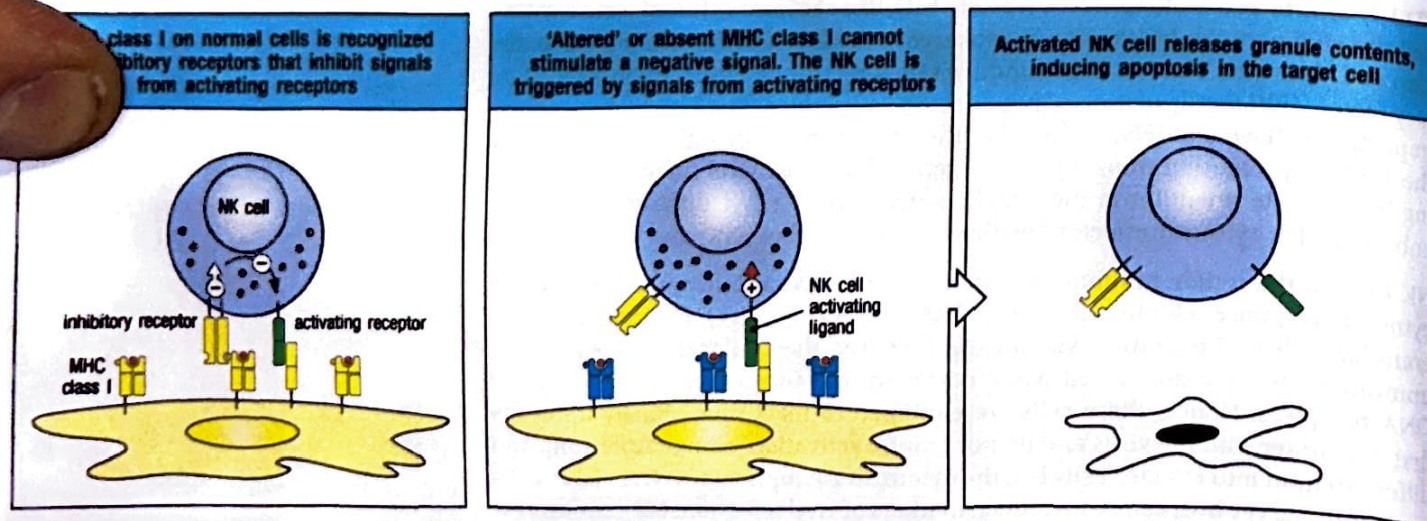


Fig. 13.2 Receptor-ligand interactions in the activation of NK cells. NK cells carry both activating and inhibitory receptors on their surface. Left panel: when an inhibitory receptor recognizes an MHC class I molecule, it sends an inhibitory signal that prevents activation via the activating receptors (which recognize

a variety of carbohydrate ligands on cell surfaces). Center panel: altered or absent MHC does not engage the inhibitory receptors, however, in those circumstances, the activating signal dominates. Right panel: as a result, cells expressing low levels of MHC molecules are susceptible to killing by NK cells.

In this case, an immunodeficiency reduced the immune system's ability to control infection by Epstein-Barr virus (EBV). EBV infects epithelial cells and B cells and is present in small numbers in most humans over the age of 15 years. Its expansion is usually well controlled by cytotoxic T cells and NK cells. A primary EBV infection triggers the activation and cell division of B cells infected by the virus. The infected B cell expresses a number of viral antigens that are targets for specific cytotoxic responses by NK cells and CD8 T cells that keep the proliferation of infected B cells under control. In most people, EBV infection remains asymptomatic or may cause infectious mononucleosis, which subsides within a span of 6–10 weeks. The normal course of acute infectious mononucleosis is described in Case 42, and you may find it helpful to read that case before embarking on this one. After resolution of the acute infection, the virus persists in a latent form in B cells, salivary glands, and epithelial cells of the nose and throat and can be shed in saliva. Occasional reactivation of the virus later in life is rapidly brought under control by EBV-specific memory cytotoxic T cells. This cellular immune surveillance is critical in maintaining the balance between host and virus. Primary and acquired deficiencies of T cell function are associated with a marked susceptibility to lethal EBV infection.

The case of Nicholas Nickleby: inefficient killing of EBV-infected B cells by cytotoxic lymphocytes.

Nicholas was brought to the pediatrician at 5 years of age because of several days of fever (38–39°C). He had no cough, runny nose, rash, diarrhea, or any other symptoms of infection. The physical examination revealed only some mildly enlarged lymph nodes in his neck, and his parents were advised to treat the fever with acetaminophen. Over the following weeks, the fevers persisted and Nicholas seemed less energetic than usual. He was brought to the doctor several times but the only consistent finding was persistent enlarged, nontender lymph nodes. Finally, after 6 weeks of illness, Nicholas complained of abdominal pain and was referred to the Children's Hospital.

Nicholas's medical history was notable for persistent and recurrent middle ear infections (otitis media) as well as several episodes of bacterial pneumonia between 2 and 3 years of age. Immunological evaluation at the time revealed decreased blood levels of IgG of 314 mg dL⁻¹ (normal 600–1500 mg dL⁻¹), and normal IgA and IgM. Nicholas was briefly treated with prophylactic antibiotics with a good response. These were discontinued at the age of 4 years, and he had no further infections or follow-up tests before his admission to Children's Hospital. The family history was notable for the presence of a maternal uncle who died of aplastic anemia following an acute febrile episode. Furthermore, Nicholas's maternal grandfather had died with a history of recurrent lymphomas.

On admission to hospital, Nicholas appeared tired but not acutely ill. His temperature was 38.5°C and his heart rate, respiration, and blood pressure were all normal. His height and weight were both in the 25th percentile for age. A few scattered small skin hemorrhages (petechiae) were noted on his legs and feet. Several lymph nodes were palpable in his neck, and these appeared significantly larger than on previous examinations. Supraclavicular, axillary, or inguinal lymph nodes were not enlarged. The tonsils were moderately enlarged but were not red, and there was no evidence of inflammation. The heart sounds were normal. The abdomen was moderately distended but soft, and slightly tender in the right upper quadrant. The liver was enlarged, and its edge was palpable 4 cm below the right costal margin.

Laboratory evaluation showed a mild anemia with a hematocrit of 28% (normal 35–40%). The white blood cell count was 6400 μL^{-1} (normal 5000–10,000 μL^{-1}). The

5-year-old child with
persistent unexplained fever
and lymphadenopathy.

History of recurrent otitis
medias and pneumonias.

High proportion of atypical
lymphocytes in blood count,
elevated ferritin, and
hemophagocytosis in the bone
marrow.

Diagnostic Criteria for HLH

A molecular/genetic diagnosis consistent with HLH.

OR clinical presentation with 5 of the following 8 criteria:

1. Fever
2. Splenomegaly
3. Cytopenias affecting at least 2 cell lines
Hgb $<90 \text{ gm l}^{-1}$
Platelets $<100,000 \mu\text{m}^{-1}$
Neutrophils $<1,000 \mu\text{m}^{-1}$
4. Hypertriglyceridemia $\leq 265 \text{ mg dl}^{-1}$
and/or
Hypofibrinogenemia $\leq 150 \text{ mg dl}^{-1}$
5. Hemophagocytosis in the bone marrow, spleen, or lymph nodes with no evidence of malignancy
6. Low or absent NK cell activity
7. Ferritin $\leq 500 \text{ ng ml}^{-1}$
8. Soluble CD25/IL-2 receptor $>2400 \text{ U ml}^{-1}$

Tests for acute EBV infection
positive.

Diagnosis of XLP in view of
family history and clinical
presentation.

Fig. 13.3 Diagnostic criteria for HLH. Failure to clear the EBV viruses leads to an ongoing immune stimulus that results in production of inflammatory cytokines (IFN γ , TNF α , IL-1, IL-2, and IL-6) and activation of lymphocytes and macrophages. The abundance of these cytokines and over-activation of lymphocytes and macrophages lead to the clinical symptoms characteristic of HLH, including fevers, cytopenias, coagulopathy, elevated ferritin, increased triglycerides, and hemophagocytosis.

platelet count was low at $47,000 \mu\text{l}^{-1}$ (normal $150,000\text{--}300,000 \mu\text{l}^{-1}$). The count of different types of white blood cells was remarkable for the very high proportion (22%) of atypical lymphocytes (normal less than 2%) (see Case 42). Liver function tests indicated significant liver damage with elevated ALT (alanine aminotransferase) at 1250 U l^{-1} (normal $5\text{--}45 \text{ U l}^{-1}$) and AST (aspartate aminotransferase) at 824 U l^{-1} (normal $5\text{--}45 \text{ U l}^{-1}$). Ferritin levels and triglycerides were also markedly elevated at $10,000 \text{ ng ml}^{-1}$ and 500 mg dl^{-1} , respectively, while the fibrinogen level was low at 75 mg dl^{-1} . A bone marrow biopsy showed hemophagocytosis (engulfment of red blood cells) by macrophages. Tests for antibodies against hepatitis A, B, and C viruses were negative. The titer of IgM antibody against EBV viral capsid antigen (VCA) was positive at more than 1:40. Anti-VCA IgG antibody was 1:320 and antibodies against Epstein-Barr nuclear antigen (EBNA) and early antigen (EA) were undetectable, consistent with an acute EBV infection (see Fig. 43.4). EBV viremia was detected in Nicholas's blood by the polymerase chain reaction (PCR).

A chest X-ray showed clear lungs and a normal-sized heart, but the lymph nodes in the mediastinum were enlarged. Ultrasound examination of the abdomen revealed a significant amount of free fluid in the abdominal cavity (ascites) and an enlarged liver. An abdominal CT scan revealed marked enlargement of lymph nodes in the retroperitoneum.

In the light of the family history and laboratory evidence of acute EBV infection, a diagnosis was made of X-linked lymphoproliferative syndrome (XLP) with fulminant infectious mononucleosis resulting in hemophagocytic lymphohistiocytosis (HLH) (Fig. 13.3). Nicholas was kept in the hospital and initially treated with an antiviral agent, ganciclovir, and intravenous immune globulin (IVIG) in an attempt to control the EBV infection. However, the fever persisted, and his liver dysfunction and ascites rapidly worsened. The glucocorticoid dexamethasone was added to his therapy along with rituximab, a monoclonal antibody to CD20 that depletes B cells containing the EBV virus. Despite these aggressive interventions, Nicholas developed severe shock symptoms, resembling those after bloodstream infection (sepsis), with diffuse vascular leakage, a fall in blood pressure (hypotension), poor circulation, and multi-organ failure. All cultures for bacterial pathogens were negative. He died 10 days after admission to the hospital.

A post-mortem examination showed that the liver was markedly enlarged. Fluid had accumulated in the abdomen and around the lungs (pleural effusions). EBV was identified by culture and PCR in the liver and bone marrow. There was a striking infiltration of the liver, spleen, and lymph nodes by a mixed population of mononuclear cells including small lymphocytes, plasma cells, and lymphoblasts. In the liver, these infiltrates were associated with extensive tissue death (necrosis). Examination of the bone marrow revealed a decreased number of erythroid, megakaryocytic, and myeloid cells, along with increased numbers of histiocytic cells, lymphocytes, and plasma cells.

To confirm a diagnosis of X-linked lymphoproliferative syndrome, expression of SAP (SH2D1A) mRNA was analyzed by Northern blotting, revealing a complete absence of the product (Fig. 13.4). None of the four exons encoding SAP could be amplified by PCR, consistent with a complete deletion of the gene.

Fig. 13.4 Northern blot analysis of SAP mRNA in peripheral blood mononuclear cells obtained from Nicholas. RNA was prepared from the Jurkat T cell line (Jk) and from the peripheral blood mononuclear cells of a control (C) and the patient (P). A ^{32}P -labeled probe specific for SAP was used to probe the Northern blot, and a β -actin probe was used as a loading control. Courtesy of Hans Oettgen.

X-linked lymphoproliferative disease (XLP).

The child in this case was diagnosed with X-linked lymphoproliferative syndrome, which is characterized by the clinical triad of vulnerability to EBV infections, lymphoma, and dysgammaglobulinemia. In 60-70% of cases, XLP is caused by loss of function of the signaling lymphocytic activation molecule (SLAM)-associated protein (SAP), which is encoded by the *SH2D1A* gene. The condition is inherited through unaffected females, an indication that the susceptibility trait is due to a gene on the X chromosome.

SAP is a cytosolic protein expressed in T, NKT, and NK cells, all of which are important for clearing viral infections. The SAP protein acts primarily as an adaptor molecule for the SLAM family of receptors, and mediates SLAM signaling. SAP consists almost exclusively of an Src homology 2 (SH2) domain, which connects to other proteins by tyrosine-phosphorylated motifs. SLAM receptors contain an extracellular Ig-like domain and a cytoplasmic tail with immunoreceptor tyrosine-based switch motifs (ITSM) that act as a docking site for SAP. While most SH2 domain-containing proteins bind to ITSM through two contact sites, the SH2 domain of SAP binds to the ITSM of SLAM at 3 contact sites, accounting for the tight association between SAP and SLAM. The SAP SH2 domain has also the unique ability to bind simultaneously to SLAM via interaction with the ITSM motif and to the SH3 domain of the src family kinase FynT using Arginine 78. Once SAP is bound to SLAM, a conformational change in SAP allows it to recruit FynT. FynT then mediates intracellular signaling and activation. SAP also blocks inhibitory signals to SLAM from other SH2 domain-containing protein phosphatases such as SH2 domain-containing tyrosine phosphatase-1 and -2 (SHP-1; SHP-2) and SH2 domain-containing inositol phosphatase (SHIP-1). It is speculated that SAP's high affinity binding to SLAM competitively inhibits binding of these proteins (Fig. 13.5).

Therefore, loss of function of SAP results in reduced activity of the SLAM family of receptors, causing pleomorphic effects on the immune system. SLAM receptors are found on dendritic cells, macrophages, NK cells, and B and T lymphocytes. In CD8 T cells, activation of SLAM receptors augments TCR signaling, T cell proliferation, IFN γ secretion, and cytotoxic granule release. SAP expression and SLAM receptor signaling are also important in CD4 $^{+}$ follicular helper T (T_{FH}) cells. T_{FH} cells express CXCR5 and migrate to the T cell zone of the B cell follicle where they promote germinal center (GC) formation, isotype switching, and B cell responses to protein antigens. SAP-deficiency is thought to interfere with the ability of T_{FH} to stay in the GC and form stable T-B cell contacts. This defect in T_{FH} may be mediated by reduced expression of inducible T cell costimulator (ICOS) following TCR stimulation in SAP-deficient T cells. In addition to SAP and SLAM function in CTLs and T_{FH} cells, SLAM receptors may also regulate excessive T cell activation through restimulation-induced cell death (RICD). During an exuberant immune response characterized by excessive pro-inflammatory cytokines, strong T cell stimulation can result in apoptosis or RICD. RICD acts as a set of breaks on the immune system to prevent excessive inflammation and lymphoproliferation. Stimulation of the SLAM receptor NTB-A is thought to partially mediate this process. SLAM receptor signaling is not only important in T cells but is also vital for NK cell function. Similar to CD8 T cells, SLAM receptor engagement in NK cells promotes IFN γ secretion and cytotoxicity. SLAM receptor family signaling is also important for the

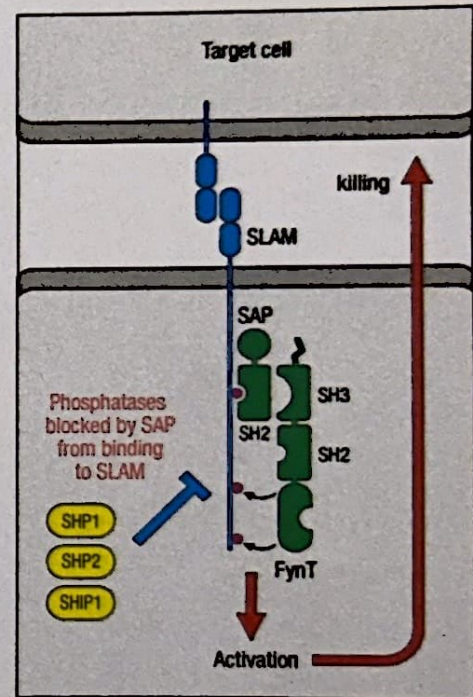
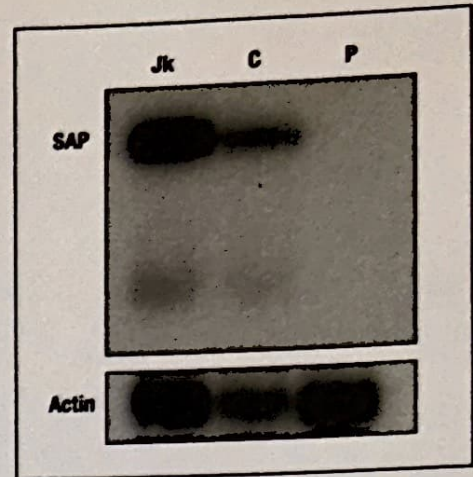


Fig. 13.5 SAP-mediated SLAM receptor signaling. SAP activates SLAM receptors by two mechanisms. The SH2 domain of SAP binds the immunoreceptor tyrosine-based switch motif (ITSM) of SLAM in a three-pronged manner. SAP then recruits FynT to the receptor complex through binding of the SH3 domain of FynT to arginine 78 within SAP. FynT then initiates a cascade of activating signals that lead to enhance cytotoxicity. Second, SAP prevents the binding to SLAM receptors of inhibitory proteins such as SH2 domain-containing tyrosine phosphatase-1 and -2 (SHP-1; SHP-2), and SH2 domain-containing inositol phosphatase (SHIP-1).

development of premature NKT cells in the thymus and loss of SAP function, and SLAM activation leads to reduced numbers of NKT cells in XLP patients.

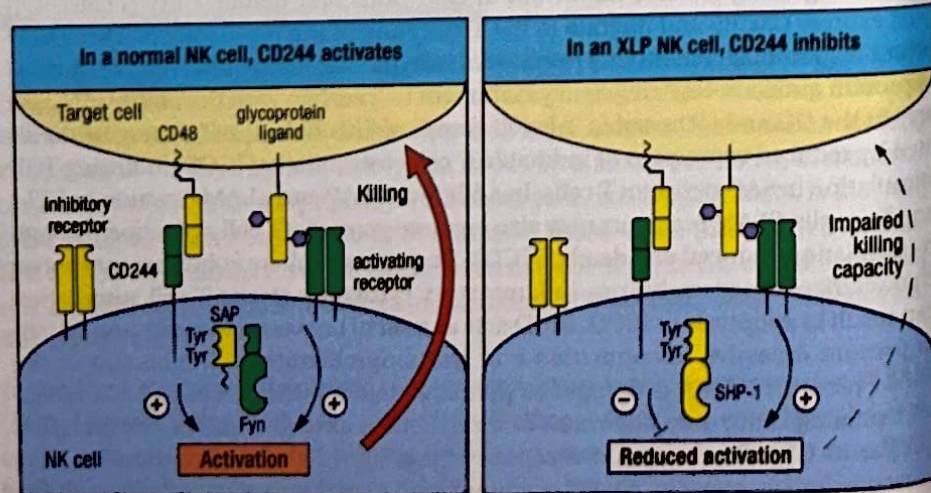
The multiple immune defects that result from SAP deficiency and abnormal SLAM signaling are thought to cumulatively result in the XLP phenotype characterized by fulminant EBV infections, hypogammaglobulinemia, and lymphoma.

The vulnerability of these patients to EBV is partially explained by the defective cytotoxic responses of CD8 T and NK cells, resulting in the persistence of EBV-infected cells. In particular, B cells in healthy individuals who are infected with EBV increase expression of the SLAM family member CD48. CD48 interacts with another SLAM family member on NK cells, CD244, providing an activating signal that enables the NK cells to kill the infected B cells (Fig. 13.6). Signaling via CD244 is critical in driving NK killing of EBV-infected target cells. Defects in SLAM signaling due to the SAP deficiency in XLP prevents this process, allowing EBV to evade the immune system. Additionally, NKT cells, which are likely important in antiviral responses, are reduced in XLP patients and likely contribute to the ongoing EBV infection.

Failure to clear the EBV viruses coupled with defective RICD creates an ongoing immune stimulus, triggering massive over-proliferation of lymphocytes (lymphoproliferation), activation of macrophages, and generation of a cytokine storm or excessive production of inflammatory cytokines such as IFN γ , TNF α , IL-1, IL-2, and IL-6. The overabundance of these cytokines leads to the clinical symptoms characteristic of fulminant EBV infection that can progress to HLH, including fevers, cytopenias, and coagulopathy. The abundant T cell-derived cytokines, particularly IFN γ , also trigger the activation of monocytes and macrophages, which leads to overproduction of ferritin. The activated macrophages also engage in indiscriminate phagocytosis of surrounding cells. Histologic analysis of tissues from patients with fulminant infectious mononucleosis often reveals hemophagocytosis, a phenomenon in which some macrophages seem to engulf entire red blood cells. Activated CD8 T cells, B cells, and macrophages infiltrate affected tissues including the liver, bone marrow, spleen, and lymph nodes. Ultimately, patients can go on to develop seizures, liver failure, enhanced vascular permeability, and loss of intravascular volume leading to hypotension. This is analogous to the clinical scenario observed in toxic shock syndrome (see Case 42), in which uncontrolled T cell activation and cytokine secretion can also lead to multi-organ damage and a clinical picture resembling sepsis. Unfortunately, the fulminant EBV/HLH that develops in XLP patients is often lethal. In one study of 161 boys with XLP, 57% died of complications from EBV infections.

Although more than half of the patients with XLP are extremely susceptible to EBV, one-third of the patients develop dysgammaglobulinemia without an episode of severe mononucleosis. Even in patients who ultimately develop severe EBV infections, the dysgammaglobulinemia can be the initial presentation of XLP, as was the

Fig. 13.6 Loss of SAP in NK cells converts an activating signal to an inhibitory signal for killing of EBV infected B cell. The SLAM family molecule CD244 on NK cells recognizes the SLAM family member CD48 on EBV infected B cells. CD244 signaling via SAP activates Fyn and synergizes with signals delivered by NK cell receptors to promote NK cell killing. In the absence of SAP, the tyrosine phosphatase SHP-1 is recruited and activated by CD244, and inhibits NK cell receptor signaling. This results in impaired NK cell killing.



case for Nicholas. SAP-deficient patients tend to have normal numbers of B cells; however, the proportion of memory B cells expressing IgG is low. These findings are not thought to be due to an intrinsic B cell defect. Instead, the known abnormalities in SAP-deficient T_{FH} are likely responsible for the reduced germinal center formation, isotype switching, and B cell maturation in these patients.

Lymphoma develops in 30% of patients with XLP, as it did in Nicholas's maternal grandfather. Most lymphomas occur because of B-cell transformation by EBV, and the classic t(8;14) chromosomal translocation of EBV-induced Burkitt's lymphoma has been observed in some patients with XLP. In some cases, malignancy develops without evidence of a prior EBV infection, and the cause of the lymphoma in these patients is not completely understood. Decreased immunosurveillance by NK and CD8 T cells is one possible mechanism. Additionally, the NKT cell deficiency in XLP may play a role in impaired tumor cell recognition and clearance. Finally, the reduced apoptosis of T cells that has been described in XLP may also promote the development of malignancy.

In rare instances, patients with XLP can develop other manifestations of the disease including aplastic anemia, as in Nicholas's maternal uncle, hemolytic uremic syndrome, and autoimmunity.

A minority of male patients with XLP, about 30–40%, does not carry mutations in SAP but has mutations in the *BIRC4* gene, encoding the X-linked inhibitor of apoptosis protein (XIAP). The terms XLP-1 and XLP-2 are used to distinguish patients with XLP due to SAP or XIAP mutations, respectively. Similar to patients with XLP-1, males with XIAP deficiency often present with fulminant EBV infections and hypogammaglobulinemia. Both groups of patients have NKT cell defects with absent NKT cells in XLP-1 patients and low NKT cells in XLP-2 patients. Interestingly, the increased lymphoma risk has not been documented in XIAP-deficient patients. Instead, XLP-2 patients tend to develop refractory and severe colitis that mediates much of the mortality associated with this condition. The pathophysiology of the XLP phenotype in XIAP-deficient patients is not entirely clear. XIAP is expressed more broadly than SAP and is found in both immune and non-immune cells. XIAP contains three baculoviral IAP repeat domains as well as a ring domain that acts as an E3 ubiquitin ligase. XIAP inhibits apoptosis by blocking the effects of caspases 3, 7, and 9. It can also activate the mitogen-activated protein kinase and nuclear factor- κ B (NF κ B) pathways. Additionally, XIAP is necessary for signaling through the nucleotide-binding and oligomerization domain (NOD) 2. Interestingly, NOD2 is an intracellular pattern recognition receptor that activates the NF κ B pathway; mutations in this protein are associated with Crohn's Disease (see Case 37), possibly explaining the colitis found in XLP-2 patients.

Recently, several novel immunodeficiencies have been identified with features of EBV-lymphoproliferation, including mutations in a third X-linked gene, magnesium transporter 1 (*MAGT1*). Female patients can also present with an XLP-like phenotype, leading to speculation that autosomal gene mutations may also cause the disease. Newly described deficiencies in IL-2-inducible T cell kinase (ITK), serine/threonine stress kinase 4 (STK4), CD27, and coronin 1A have confirmed this hypothesis. Of these, the ITK deficiency is probably the best characterized with case reports on 9 patients. The clinical course is characterized by EBV lymphoproliferative disorder that often culminates in HLH. Lymphoma is common in these patients along with autoimmune cytopenias, nephritis, and thyroiditis. Pulmonary involvement is a distinctive feature of ITK deficiency. Patients also present with hypogammaglobulinemia, progressive CD4⁺ lymphopenia, and low NKT cells. The function of ITK is well known. ITK is recruited to the cell membrane after TCR stimulation and activated by lymphocyte specific protein tyrosine kinase (LCK). In turn, ITK phosphorylates and activates phospholipase C γ 1 (PLC γ 1) leading to downstream TCR signaling transduction, including release of calcium from the endoplasmic reticulum and translocation of nuclear activator of T cells (NFAT) into the nucleus. How defects in ITK cause EBV lymphoproliferation is less well understood and requires further investigation.

Questions.

- 1 Did Nicholas's history provide any clues to his diagnosis before his presentation with fulminant infectious mononucleosis?
- 2 How can the diagnosis of X-linked lymphoproliferative syndrome be made?
- 3 If a girl presents with fulminant infectious mononucleosis and evidence of acute EBV infection, what might be the basis of her disease?
- 4 Nicholas has a 3-year-old brother, Alexander, who has SAP deficiency. What can be done to prevent the occurrence of disease in this boy?